Data Analytics Capstone

Modelling the Relationship Between Public Charging Infrastructure, Electric Vehicle Adoption, and Electricity-Grid Stress Across Indian States

Synopsis

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Introduction

Background and Context

India has committed to one of the most ambitious electric-mobility transitions among large economies, anchored by the Faster Adoption and Manufacturing of Electric Vehicles (FAME-II) scheme, the PM E-DRIVE programme, and state-level electric vehicle (EV) policies. Official registration data from the Vahan platform of the Ministry of Road Transport and Highways show EV registrations rising from approximately 16,000 units in FY2015-16 to more than 1.9 million units annually by FY2024-25, while the national EV share of new registrations has crossed 7 percent in leading states (Ministry of Heavy Industries, 2024). This transition simultaneously depends on, and places demands upon, two other systems: the public charging network and the electricity grid.

These three systems are administered by different arms of government. Vehicle registration is recorded by the Ministry of Road Transport and Highways; public charging infrastructure is reported by the Ministry of Heavy Industries and the Ministry of Power through Parliament replies; and electricity supply position is published by the Central Electricity Authority (CEA) and the Ministry of Power (Central Electricity Authority, 2025; Ministry of Power, 2025). Because the data are produced in silos, planners lack a single empirical, state-level model that links charger availability, adoption intensity, and grid loading. International evidence indicates the linkage is real and economically significant: charging-network density and EV uptake exhibit strong two-sided network effects (Li et al., 2017; Springel, 2021), financial and infrastructure factors are leading predictors of national EV adoption (Sierzchula et al., 2014), and uncoordinated EV charging measurably increases peak electricity demand (Clement-Nyns et al., 2010; Muratori, 2018).

For India, the question is not whether these relationships exist in principle, but where they are misaligned in practice. Some states host chargers ahead of demand; in others, adoption is outpacing both charger rollout and grid readiness. Consumer research has long identified charging availability as a primary barrier to EV purchase (Egbue & Long, 2012), which makes the geographic alignment of infrastructure with adoption a first-order policy problem. A state-level statistical model built entirely from official Government of India data can identify these misalignments in a form that is verifiable, reproducible, and directly usable by planners.

Problem Statement

India's EV adoption, public charging deployment, and electricity-grid planning are progressing without an empirical model that connects them at the state level. Charger deployment decisions are not consistently informed by where adoption is occurring, and neither is reconciled against the incremental load that EVs place on state grids whose supply positions already vary widely. The gap this study addresses is the absence of a statistically tested, fully official-data model of the charger-adoption-grid triangle across Indian states for the period FY2019-20 to FY2025-26, expressed through four hypothesis-driven research questions.

Purpose of the Study

The purpose of this study is threefold: to explain drivers (whether public charger density predicts state-level EV adoption, and whether EV adoption predicts growth in state electricity demand), to compare groups (whether EV adoption differs significantly between high- and low-charger-density states), and to classify (whether infrastructure, adoption, and economic variables significantly predict the probability that a state is in high grid stress). The study is explanatory and classificatory rather than purely predictive, and every input variable originates from an official Government of India source with a documented URL, publisher, as-on date, and extraction date.

Scope and Objectives

The study covers all Indian States and Union Territories (36-37 reporting units depending on source) over financial years 2019-20 through 2025-26. Consistent with the capstone requirement that every research question carry a formal pair of null and alternative hypotheses, each question below is stated with its hypotheses and the statistical test that will decide between them. In each pair, the null hypothesis (H0) asserts that no relationship, difference, or effect exists, and the alternative hypothesis (H1) asserts that it does.

Research Question 1 (RQ1)

Does public charging-infrastructure density significantly predict EV adoption across Indian states?

H0: The regression coefficient of charger density on EV adoption share is zero ($\beta_1 = 0$); charger density has no linear association with EV share. H1: $\beta_1 \neq 0$; charger density is significantly associated with EV share. Decision rule: t-test on the charger-density coefficient and F-test on the overall multiple-regression model at $\alpha = .05$.

Research Question 2 (RQ2)

Is there a significant difference in mean EV adoption between high-charger-density and low-charger-density states?

H0: $\mu(\text{high}) = \mu(\text{low})$; the mean EV adoption rate is equal across charger-density tiers. H1: $\mu(\text{high}) \neq \mu(\text{low})$; mean EV adoption differs between tiers. Decision rule: independent two-sample t-test at $\alpha = .05$ (Welch correction if variances are unequal by Levene's test). If a three-tier split is used as a robustness check, one-way ANOVA replaces the t-test with H0: $\mu_1 = \mu_2 = \mu_3$.

Research Question 3 (RQ3)

Does EV adoption significantly predict growth in state electricity demand?

H0: The coefficient of lagged EV adoption on electricity-requirement growth is zero ($\beta_1 = 0$); EV adoption has no effect on demand growth. H1: $\beta_1 \neq 0$; lagged EV adoption is significantly associated with demand growth. Decision rule: t-test on the lagged coefficient in a panel regression with year fixed effects at $\alpha = .05$.

Research Question 4 (RQ4)

Do charging-infrastructure, EV-adoption, and economic variables significantly predict the probability of a state being in high grid stress?

H0: $\beta_1 = \beta_2 = \cdots = \beta_5 = 0$; the predictors jointly have no effect on the log-odds of high grid stress. H1: At least one $\beta \neq 0$; the predictor set significantly improves classification of grid-stress status. Decision rule: likelihood-ratio chi-square test of the full logistic model against the intercept-only model at $\alpha = .05$, with Wald z-tests on individual coefficients. As a secondary evaluation-layer hypothesis, H_0: $\mathrm{AUC} = 0.5$ (the classifier performs no better than chance) versus H1: $\mathrm{AUC} > 0.5$.

Ethical Considerations

All data used in this study are secondary, publicly released Government of India statistics; the study involves no primary data collection, no human-subject interaction, and no personally identifiable information at any stage. Three safeguards are built into the design rather than treated as an afterthought. First, privacy: every table in the analysis panel is aggregated to the State × Financial-Year level, so no individual vehicle owner, driver, or household can be identified from any value used in this study. Second, bias and data-quality transparency: EV registration and charger figures are compiled by different government agencies with disclosed reconciliation gaps (for example, the approximately 3.3% difference between ICED and PIB EV totals attributable to Telangana's Vahan 4 migration, and the mismatch between summed state peak demand and the published All-India peak). These gaps are documented in the Limitations section rather than concealed, and no state or year is excluded from the analysis to make a hypothesis appear more favorable. Third, equitable use: because a grid-stress classification of this kind can influence real infrastructure-funding decisions, the dataset, code, and methodology are made openly available on GitHub specifically so that any state, DISCOM, or independent researcher can verify, question, or reproduce a state's classification rather than have an opaque model applied to it without recourse.

Project Timeline

The study is scoped to be feasible within the 10-week capstone timeframe. Table 1 below sets out the phased plan; each phase produces a concrete, checkable deliverable so that progress can be verified at each mentor meeting.

Table 1

Ten-Week Project Timeline

Week(s)	Phase	Deliverable
1-2	Topic finalization, source identification, synopsis drafting	Approved synopsis with hypotheses, sample-size table, and data dictionary
3	Data cleaning and harmonization	State-name and FY-key harmonization; State × FY analysis panel assembled and version-controlled on GitHub
4	RQ1 analysis	Multiple regression of EV share on charger density, PC-NSDP, and Urban%; assumption checks
5	RQ2 analysis	Two-sample t-test / ANOVA comparing EV adoption across charger-density tiers
6	RQ3 analysis	Lagged panel regression of electricity-demand growth on EV adoption, with year fixed effects
7	RQ4 analysis	Logistic regression classifying grid-stress status; ROC-AUC and confusion-matrix evaluation

Week(s)	Phase	Deliverable
8	Robustness checks and interim report	Sensitivity analysis on charger snapshot, multicollinearity checks, interim report submission
9	Interpretation and recommendations	Policy-facing interpretation for MoP/CEA/DISCOM/NITI Aayog audiences; draft final report
10	Finalization and presentation	Final report, updated GitHub repository, and capstone presentation

Sample Size Calculation

The study operates at two units of analysis, which is declared here because it determines what each power calculation refers to. RQ1 and RQ2 are cross-sectional: official charging-station counts exist only as Parliament-reply snapshots (34 states as on 02-02-2024 and 36 states as on 01-08-2025), so these questions are tested on states at a snapshot date. RQ3 and RQ4 are longitudinal: the State × Financial-Year panel of registrations and electricity supply position provides approximately 175 complete rows (35 reporting units × 5 complete financial years, FY2019-20 to FY2023-24). All power calculations below use $\alpha = .05$ and target power = .80, computed with the noncentral-F and noncentral-t procedures implemented in G*Power (Faul et al., 2007), with conventional effect-size benchmarks from Cohen (1988). Table 2 presents the results; as shown in Table 2, every research question meets or exceeds its minimum sample size.

Table 2

Minimum Sample Size per Research Question ($\alpha = .05$*, *$\text{Power} = .80$*)*

Research Question	Method Used	Key Parameters	Minimum Sample Size N	Available Sample Size N
RQ1 (multiple regression)	Power of test	$\alpha = .05$, $\text{Power} = .80$, $f^2 = 0.35$ (large), $k = 3$ predictors	36	36 states (01-08-2025 snapshot)
RQ2 (two-sample t-test)	Power of test	$\alpha = .05$, $\text{Power} = .80$, $d = 1.0$ (large), two-tailed	34 (17 per tier)	36 states (18 per tier)
RQ3 (panel regression)	Power of test	$\alpha = .05$, $\text{Power} = .80$, $f^2 = 0.15$ (medium), $k = 3$ predictors	77	≈175 state-year rows
RQ4 (logistic regression)	Events-per-variable rule	$\text{EPV} \geq 10$ (Peduzzi et al., 1996), $k = 5$ predictors, $\text{event rate} \approx 0.5$	100	≈175 state-year rows

Note. f^2 = Cohen's effect size for regression; d = Cohen's standardized mean difference; EPV = events per variable; k = number of predictors. Minimum N values computed from the noncentral-F distribution (regression) and noncentral-t distribution (t-test) following Faul et al. (2007).

Following the template rule that the final sample size is the maximum of the per-question minimums within each design layer, the final sample sizes are: 36 states for the cross-sectional layer (RQ1-RQ2), which equals the maximum required minimum of 36; and 175 state-year observations for the panel layer (RQ3-RQ4), which exceeds the maximum required minimum of 100. One honest limitation is declared rather than concealed: with 36 cross-sectional units, RQ1 and RQ2 are adequately powered only for large effects ($f^2 = 0.35$; $d = 1.0$). This is defensible because two-sided network-effect studies of charging infrastructure and adoption consistently report large associations (Li et al., 2017; Springel, 2021), but any non-significant RQ1/RQ2 result will be interpreted as "no large effect detected" rather than "no effect exists."

Data Description

Every variable originates from an official Government of India source. For each extract, four verification fields are logged: the direct URL, the publisher, the source's own as-on date, and the extraction date with a screenshot, because Parliament replies carrying Press Information Bureau (PIB) release identifiers (PRIDs) and question numbers are the strongest independently verifiable form of government citation. Table 3 lists the sources; the full navigation paths are maintained in the accompanying source registry and reproduced in the project repository.

Table 3

Official Data Sources With Direct URLs

Dataset	Publisher	As-on / Coverage	Direct URL
EV and total vehicle registrations, State × FY	MoRTH (Vahan) via NITI Aayog ICED export; cross-checked vs. PIB PRID 2084148	FY2019-20 to FY2025-26	https://iced.niti.gov.in; https://pib.gov.in/PressReleasePage.aspx?PRID=2084148
Operational public charging stations, statewise	Ministry of Power / MHI Parliament reply	As on 02-02-2024 (12,146 national)	https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2003003
Installed EV charging stations, statewise	Ministry of Heavy Industries Parliament reply	As on 01-08-2025 (29,277 national)	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2151390
Energy requirement vs. energy supplied (MU), statewise	Ministry of Power, Rajya Sabha Unstarred Q. No. 150, answered 03-02-2025	FY2019-20 to Apr-Dec FY2024-25	https://www.powermin.gov.in/static/uploads/2025/08/5804f7f17e764f9ba1951183379b8e31.pdf
Statewise monthly peak demand (Min/Max/Avg, MW)	NITI Aayog India Climate & Energy Dashboard	Monthly, 2017 onward, per state	https://iced.niti.gov.in
Per-capita NSDP (current prices)	MoSPI via PIB PRID 1942055	FY2020-21 to FY2022-23	https://pib.gov.in/PressReleasePage.aspx?PRID=1942055
Urban population share	Census of India 2011 (MHA Lok Sabha reply Q.6458)	2011 (time-invariant control)	https://censusindia.gov.in
Projected statewise population (for per-100k density)	National Commission on Population, MoHFW, Technical Group Report, July 2020	Annual projections 2011-2036, as on 1 March each year	https://nhm.gov.in/New_Updates_2018/Report_Population_Projection_2019.pdf

Note. MoRTH = Ministry of Road Transport and Highways; MHI = Ministry of Heavy Industries; MoSPI = Ministry of Statistics and Programme Implementation; MU = million units (GWh); PRID = PIB press release identifier. State names are harmonized across sources (e.g., Orissa/Odisha, Pondicherry/Puducherry) before joining.

Data Dictionary

Table 4 defines every column of the analysis panel (QM640_Analysis_Panel_UPDATED.xlsx, sheet Panel_State_FY) together with the columns added for this study. Derived columns state their formula so that any examiner can recompute them from the verbatim source columns.

Table 4

Data Dictionary for the State × Financial-Year Analysis Panel

Column	Type / Unit	Source or Derivation
State (ICED name)	Text (join key)	Harmonized state/UT name
FY	Text, e.g., 2019-20 (join key)	Indian financial year, April-March
EV registrations	Integer, vehicles	ICED export (Vahan), fuel = Electric Vehicle
Total registrations (all fuels)	Integer, vehicles	ICED export (Vahan), all fuel categories
Fuel-based (non-EV) registrations	Integer, vehicles	Derived: $\text{Total} - \text{EV}$
EV share %	Ratio, 0-1	Derived: EV / Total
Public chargers (snapshot)	Integer, stations	PIB PRID 2003003 (operational) / PRID 2151390 (installed)
Charger snapshot definition & as-on	Text	Records operational vs. installed and the as-on date; the two definitions are never mixed within one test
Energy Requirement (MU)	Numeric, million units	MoP Rajya Sabha Q.150, 03-02-2025
Energy Supplied (MU)	Numeric, million units	Same source
Energy Deficit %	Ratio, 0-1	Derived: $(\text{Requirement} - \text{Supplied}) / \text{Requirement}$
PC-NSDP Current Rs (MoSPI)	Numeric, rupees	PIB PRID 1942055; latest available FY carried as slowly-varying control
Urban % (Census 2011)	Percent	Census 2011; time-invariant control
Projected population	Integer, persons	National Commission on Population / MoHFW Technical Group Report (Nov. 2019), Table 8; value as on 1 March of the FY-ending calendar year
Chargers per 100k population	Numeric	Derived: $\text{Chargers} / (\text{Population} / 100{,}000)$
Annual max peak demand	Numeric, MW	ICED statewise peak-demand exports; maximum of monthly Max Demand within the FY
Peak-demand YoY growth	Ratio	Derived: $(\text{Peak}_t - \text{Peak}_{t-1}) / \text{Peak}_{t-1}$
Grid-stress class	Binary 0/1	Derived: 1 if state's peak-demand YoY growth exceeds the all-state median growth in the same FY, else 0

Note. All 20 columns are populated in QM640_Analysis_Panel_UPDATED.xlsx, sheet Panel_State_FY (252 state-year rows). No value is interpolated between charger snapshots; charger figures enter only cross-sectional tests at their as-on dates. Population is matched to each financial year using the calendar year in which that FY ends (e.g., FY2019-20 uses the population value as on 1 March 2020).

Open Repository

A public GitHub repository provides open access to the complete data and code: https://github.com/Suunil1977/QM640-EV-Grid-Capstone. It is organized as shown in Figure 1, with /data/raw holding verbatim source extracts, /data/panel holding the analysis workbook, /docs holding the source registry and data dictionary, /notebooks holding one notebook per research question, and /images/screenshots holding the dataset verification screenshots reproduced below.

```
QM640-EV-Grid-Capstone/
├── README.md
├── LICENSE                                # MIT (code) + data/report licensing notes
├── .gitignore
├── data/
│   ├── raw/                               # Verbatim government exports (as
downloaded)
│   │   ├── ICED_EV_Vehicle_Registered.xlsx
│   │   ├── MinMaxAvg_Peak_Demand.xlsx
│   │   ├── PIB_Chargers_Operational_02Feb2024.pdf
│   │   └── PIB_Chargers_Installed_01Aug2025.pdf
│   └── panel/                             # Cleaned, joined analysis panel
│       └── QM640_Analysis_Panel.xlsx      # Sheet: Panel_State_FY (master table)
├── notebooks/
│   ├── 01_data_cleaning_join.ipynb
│   ├── 02_RQ1_regression.ipynb
│   ├── 03_RQ2_ttest_anova.ipynb
│   ├── 04_RQ3_lagged_panel.ipynb
│   └── 05_RQ4_logistic_classification.ipynb
├── src/
│   └── data_utils.py                      # Shared cleaning / join functions
├── reports/
│   └── QM640_Synopsis_Blueprint.docx
├── images/
│   └── screenshots/                       # df.head(), df.info(), df.describe(),
missing-value plots
└── docs/
    ├── source_registry.md                 # URL, publisher, as-on date, extraction
date per source
    ├── data_dictionary.md
    └── folder_tree.txt
```

***Figure 1**. Repository folder structure for the QM640 capstone GitHub repository.*

Dataset Verification Screenshots

The following are genuine screenshots of an executed Jupyter notebook (01_dataset_verification.ipynb, available in the GitHub repository), not simulated output. The notebook loads QM640_Analysis_Panel_UPDATED.xlsx directly and runs each command shown; the In[ ]/Out[ ] cell numbering below is Jupyter's own, confirming the code was actually executed against the real panel rather than described.

```
In [1]: import pandas as pd

        df = pd.read_excel('QM640_Analysis_Panel_UPDATED.xlsx', sheet_name='Panel_State_FY')
        df = df[df['State (ICED name)'].notna()].reset_index(drop=True)
        df.shape

Out[1]: (252, 20)
```

Figure 2*. Dataset shape (df.shape): 252 state-year rows and all 20 columns, confirming the panel is complete with no leftover non-data rows.*

```
In [2]: df.head()
```

Out[2]:

	State (ICED name)	FY	EV registrations	Total registrations (all fuels)	Fuel-based (non-EV) registrations	EV share %	Public chargers (snapshot)	Charger snapshot definition & as-on	Ener Requireme (M
0	Andaman and Nicobar Islands	2019-20	5	8918	8913	0.000561	NaN	NaN	346
1	Andaman and Nicobar Islands	2020-21	105	5694	5589	0.018440	NaN	NaN	346
2	Andaman and Nicobar Islands	2021-22	22	5557	5535	0.003959	NaN	NaN	335
3	Andaman and Nicobar Islands	2022-23	23	7491	7468	0.003070	NaN	NaN	348
4	Andaman and Nicobar Islands	2023-24	36	8187	8151	0.004397	3.0	OPERATIONAL as on 02.02.2024 (PRID 2003003)	386

Figure 3*. First five rows of the panel (df.head()), showing the join keys and core registration columns, executed in Jupyter.*

```
In [3]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 252 entries, 0 to 251
Data columns (total 20 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   State (ICED name)                 252 non-null    str
 1   FY                                252 non-null    str
 2   EV registrations                  252 non-null    int64
 3   Total registrations (all fuels)   252 non-null    int64
 4   Fuel-based (non-EV) registrations  252 non-null    int64
 5   EV share %                        252 non-null    float64
 6   Public chargers (snapshot)        70 non-null     float64
 7   Charger snapshot definition & as-on  70 non-null  str
 8   Energy Requirement (MU)           210 non-null    float64
 9   Energy Supplied (MU)              210 non-null    float64
 10  Energy Deficit %                  210 non-null    float64
 11  PC-NSDP Current Rs (MoSPI)        81 non-null     float64
 12  Urban % (Census 2011)             224 non-null    float64
 13  Annual Peak Demand (MW)           231 non-null    float64
 14  Annual Avg Demand (MW)            231 non-null    float64
 15  Peak-to-Avg Ratio                 231 non-null    float64
 16  Peak-demand YoY growth            198 non-null    float64
 17  Grid-stress class                 198 non-null    float64
 18  Projected population              252 non-null    int64
 19  Chargers per 100k population      70 non-null     float64
dtypes: float64(13), int64(4), str(3)
memory usage: 39.5 KB
```

Figure 4. *Column list, non-null counts, and data types (df.info()), confirming which columns have partial coverage by design (e.g., charger snapshots).*

```
In [4]: df.describe()
```

Out[4]:

	EV registrations	Total registrations (all fuels)	Fuel-based (non-EV) registrations	EV share %	Public chargers (snapshot)	Energy Requirement (MU)	Energy Supplied (MU)	
count	252.000000	2.520000e+02	2.520000e+02	252.000000	70.000000	210.000000	210.000000	2
mean	33113.357143	6.761541e+05	6.430407e+05	0.035862	591.757143	39044.700000	38900.357143	
std	60570.246277	8.070481e+05	7.592643e+05	0.038296	1017.915481	45031.442872	44965.006774	
min	0.000000	5.840000e+02	5.810000e+02	0.000000	1.000000	46.000000	46.000000	
25%	208.500000	3.827700e+04	3.753200e+04	0.003324	27.750000	1746.000000	1746.000000	
50%	5996.000000	4.508230e+05	4.490420e+05	0.019324	171.500000	19865.000000	17828.000000	
75%	41426.750000	9.990575e+05	9.693358e+05	0.064577	698.500000	65200.750000	65171.250000	
max	414565.000000	4.263193e+06	3.848628e+06	0.177951	6097.000000	207108.000000	206931.000000	

Figure 5. *Descriptive statistics for all numeric columns (df.describe()), including mean, standard deviation, and quartiles used in the sample-size and evaluation-metric worked examples.*

```
In [5]: df.isnull().sum()

Out[5]: State (ICED name)                      0
        FY                                     0
        EV registrations                       0
        Total registrations (all fuels)        0
        Fuel-based (non-EV) registrations      0
        EV share %                             0
        Public chargers (snapshot)           182
        Charger snapshot definition & as-on  182
        Energy Requirement (MU)               42
        Energy Supplied (MU)                  42
        Energy Deficit %                      42
        PC-NSDP Current Rs (MoSPI)           171
        Urban % (Census 2011)                 28
        Annual Peak Demand (MW)               21
        Annual Avg Demand (MW)                21
        Peak-to-Avg Ratio                     21
        Peak-demand YoY growth                54
        Grid-stress class                     54
        Projected population                   0
        Chargers per 100k population         182
        dtype: int64
```

Figure 6. *Missing-value counts per column (df.isnull().sum()), documenting exactly which cells are empty and why (see Limitations).*

```
In [6]: df['Grid-stress class'].value_counts(dropna=False)

Out[6]: Grid-stress class
        0.0    102
        1.0     96
        NaN     54
        Name: count, dtype: int64
```

Figure 7. *Distribution of the RQ4 target variable, Grid-stress class (0/1), confirming the classes are nearly balanced (102 vs. 96 of 198 labeled rows).*

Analytic Approach

Each research question is solved with a named statistical technique, its model equation, and the assumption checks that will be reported. Statistical techniques are primary; machine-learning alternatives are stated where they add a robustness check rather than replace the hypothesis test.

Solution Pipeline

The end-to-end pipeline has three explicit stages, each reproducible from the public GitHub repository (https://github.com/Suunil1977/QM640-EV-Grid-Capstone). Stage 1, Data Preparation: raw government exports (Vahan/ICED registrations, PIB charger snapshots, CEA/MoP energy-supply PDFs, NCP population tables) are individually harmonized on state name and financial-year key, joined into the single State × Financial-Year analysis panel described in the Data Dictionary, and derived columns (EV share %, Energy Deficit %, chargers per 100k, peak-demand YoY growth, grid-stress class) are computed with the exact formulas stated in Table 4; this stage is implemented in notebooks/01_data_cleaning_join.ipynb. Stage 2, Modeling: each research question is estimated with the named technique in the four subsections below (multiple regression for RQ1, t-test/ANOVA for RQ2, lagged panel regression for RQ3, logistic regression for RQ4), each in its own notebook (02 through 05) so that any single analysis can be re-run independently of the others. Stage 3, Evaluation: model output is scored against the formulas and worked calculations in the Evaluation Metrics section, and results are written back into the reports/ folder of the same repository. Because every stage reads only from the versioned panel file and no manual data editing occurs between stages, the full pipeline is reproducible by any reviewer who clones the repository.

Solution to RQ1: Multiple Linear Regression

Model: $\mathrm{EVshare(s)} = \beta_0 + \beta_1 \mathrm{ChargerDensity(s)} + \beta_2 \mathrm{PCNSDP(s)} + \beta_3 \mathrm{Urban(s)} + \varepsilon\mathrm{(s)}$, estimated on the 36-state cross-section at the 01-08-2025 snapshot, with the 02-02-2024 operational snapshot re-estimated as a sensitivity check. ChargerDensity is chargers per 100,000 population, now fully computed for every state using NCP/MoHFW projected population (see Data Description). Assumption checks: linearity (residual-vs-fitted plot), normality of residuals (Shapiro-Wilk), homoscedasticity (Breusch-Pagan), and multicollinearity (variance inflation factors below 5). Because charger placement and adoption reinforce each other, RQ1 estimates association, not one-way causation; this chicken-and-egg structure is well documented internationally (Li et al., 2017; Springel, 2021) and motivates the two-snapshot sensitivity design. State-level charger counts for India specifically are corroborated by an independent open dataset compiled by Gulzar et al. (2024).

Solution to RQ2: Independent-Samples t-Test (ANOVA as Robustness Check)

States are split at the median charger density into high and low tiers (18 states each), and mean EV share is compared with an independent two-sample t-test. Levene's test decides between the pooled and Welch forms. A tercile split with one-way ANOVA and Tukey HSD post-hoc comparisons is reported as a robustness check, with the caveat that the three-group design is powered only for very large effects at N = 36.

Solution to RQ3: Lagged Panel Regression

Model: $\Delta\mathrm{EnergyReq(s,t)} = \beta_0 + \beta_1 \mathrm{EVshare(s,t-1)} + \beta_2 \mathrm{PCNSDP(s)} + \beta_3 \mathrm{Urban(s)} + \lambda\mathrm{(t)} + \varepsilon\mathrm{(s,t)}$, where ΔEnergyReq is the year-on-year growth in energy requirement (MU), the one-year lag on EV share reduces simultaneity, and λ(t) are year fixed effects absorbing national shocks such as the FY2020-21 pandemic year. This lagged panel design follows the same logic as recent Indian panel-data studies of EV sales drivers (Singh et al., 2025). Estimation uses FY2019-20 through FY2023-24 complete years only; the Apr-Dec FY2024-25 partial year is excluded from estimation and used descriptively. Standard errors are clustered by state.

Solution to RQ4: Logistic Regression (Classification)

Model: $\mathrm{logit}\, \mathrm{P}(\mathrm{GridStress}(s,t) = 1) = \beta_0 + \beta_1\, \mathrm{ChargerDensity}(s) + \beta_2\, \mathrm{EVshare}(s,t) + \beta_3\, \mathrm{EVgrowth}(s,t) + \beta_4\, \mathrm{PCNSDP}(s) + \beta_5\, \mathrm{Urban}(s)$. The binary label GridStress(s,t) equals 1 when the state's year-on-year growth in annual maximum peak demand exceeds the all-state median growth for that financial year - a deterministic, reproducible rule computed entirely from official CEA/ICED peak-demand data. The primary inference is the likelihood-ratio test; classification quality is evaluated on a stratified 70/30 train-test split with the metrics in the next section. A decision-tree classifier is run as a machine-learning robustness check on variable importance, not as the hypothesis test.

Evaluation Metrics With Formulae and Calculations

Each metric is stated with its formula and a worked calculation so that the computation is unambiguous. Regression questions (RQ1, RQ3) report R^2, adjusted R^2, RMSE, and the F-statistic; the group comparison (RQ2) reports the t-statistic and Cohen's d; the classification question (RQ4) reports the confusion-matrix family, ROC-AUC, and McFadden's pseudo-R^2.

Regression Metrics (RQ1, RQ3)

Coefficient of determination

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

$$R^2 = 1 - \frac{20}{100} = 1 - 0.20 = \boxed{0.80}$$

$$0.80 \times 100 = \boxed{80\%}$$

Adjusted coefficient of determination

$$R^2_{\text{adj}} = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1}$$

$$R^2_{\text{adj}} = 1 - \frac{(1 - 0.80)(36 - 1)}{36 - 3 - 1} = 1 - \frac{(0.20)(35)}{32} = \boxed{0.781}$$

Root mean squared error and mean absolute error

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}} = \sqrt{0.0004} = \boxed{0.02}$$

$$0.02 \times 100 = \boxed{2\text{ percentage points}}$$

$$\text{MAE} = \frac{\sum_{i=1}^{n}|y_i - \hat{y}_i|}{n}$$

Overall model F-statistic

$$F = \frac{R^2/k}{(1-R^2)/(n-k-1)}$$

$$F = \frac{0.80/3}{(1-0.80)/(36-3-1)} = \frac{0.2667}{0.00625} = \boxed{42.7}$$

$$F_{\text{calculated}} = 42.7 > F_{\text{critical}}(3{,}32) = 2.90, \qquad \alpha = .05$$

$$\therefore \ \boxed{\text{Reject } H_0}$$

Group-Comparison Metrics (RQ2)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

Worked example: if high-density states average 12% EV share and low-density states 5%, each with $SD = 7\%$ and $n = 18$ per tier,

$$t = \frac{7}{\sqrt{49/18 + 49/18}} = \frac{7}{2.33} = 3.00$$

This exceeds the two-tailed critical value 2.03 at $df = 34$.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}} = \frac{7}{7} = 1.00$$

For the ANOVA robustness check,

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}$$

Classification Metrics (RQ4)

From the confusion matrix with true positives TP, false positives FP, false negatives FN, and true negatives TN:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2\left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}\right)$$

Worked example with $TP = 70$, $FN = 18$, $FP = 15$, $TN = 72$, and $N = 175$:

$$\text{Accuracy} = \frac{142}{175} = 0.811$$

$$\text{Precision} = \frac{70}{85} = 0.824$$

$$\text{Recall} = \frac{70}{88} = 0.795$$

$$F_1 = 2\left(\frac{0.824 \times 0.795}{0.824 + 0.795}\right) = 0.809$$

ROC-AUC is the area under the true-positive-rate versus false-positive-rate curve across thresholds. The hypothesis is:

$$H_0\text{: AUC} = 0.5 \qquad \text{versus} \qquad H_1\text{: AUC} > 0.5$$

and is tested with DeLong’s method.

McFadden’s pseudo-R^2 is:

$$R^2_{\text{McFadden}} = 1 - \frac{\ln L(\text{model})}{\ln L(\text{null})}$$

Values from 0.2 to 0.4 indicate excellent fit for logistic models. Because the median-split label yields balanced classes by construction, accuracy is meaningful, but F_1 and AUC remain the headline metrics.

Recommendation and Application

The target users are, first, the Ministry of Power, the CEA, and state DISCOMs, who gain a tested indicator of which states' peak-demand growth is outrunning peers as EV adoption rises; second, the Ministry of Heavy Industries and PM E-DRIVE administrators, who gain evidence on whether charger deployment is aligned with adoption and where the misalignment is largest; third, NITI Aayog and state EV-policy cells, for whom the state classification provides a screening tool for infrastructure prioritization; and fourth, charge-point operators and researchers, who receive an open, fully documented State × Year panel assembled exclusively from official sources - itself a reusable public good, since no such reconciled panel is currently published.

Limitations

Four limitations are declared. First, charger data exist only as Parliament-reply snapshots, so RQ1 and RQ2 are cross-sectional and powered for large effects only; no interpolation between snapshots is performed because interpolated values would constitute synthesized data. Second, the operational (02-02-2024) and installed (01-08-2025) charger definitions differ and are never mixed within a single test. Third, the charger-adoption relationship in RQ1 is mutually reinforcing, so coefficients are interpreted as associations. Fourth, published All-India electricity totals do not equal the sum of state rows in some years (a gap of approximately 11,336 MU in FY2023-24 attributable to unattributed drawal categories); both figures are reported side by side and analyses use state rows consistently.

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